Lesson Plan

Teacher Name: Ms. Lubna Kakkengal and Mr. Dilan Devendra	**Class/Section:** Grade 4
Subject: TCD: Matatalab Pro Coding	**Date: Week 10 (2 Nov to 6 Nov)**
Class Description: Grade 4 TCD class, Matatalab Pro Session 1 of 4 (1 hour 30 minutes). Students refresh screen-free coding with the Command Tower and coding blocks, then use the Pro Set angle blocks to program precise turns and draw simple shapes.	
TA/LSA Role (if applicable):	

Differentiated Learning Groups (Data)			
Desert Fox	**Camel**	**Oryx**	**Falcon**

Lesson Overview		
Lesson Topic	Matatalab Pro Refresh and Angle Blocks	
Standard(s)	Math: 4.MD.C.5 (angles as turns, measuring angles in degrees); Technology: algorithms and sequencing	
Learning Objective(s) (WALT): We are learning to program precise turns using angle blocks.		**Success Criteria (WILF):** I can build a sequence of movement blocks and run it. I can explain that a full turn is 360 degrees and a quarter turn is 90 degrees. I can use an angle block to make the robot turn exactly the amount I need. I can fix my code when the robot does not do what I planned.
Literacy Strategies	Word wall with sequence, algorithm, angle, degree and debug; partners use the frame: First the robot ..., then it turns ... degrees.	
Assessment	Observation of coding with a checklist; shape challenge results; live quiz score on the TCD website.	
Targeted Vocabulary	algorithm, sequence, angle, degree, quarter turn, half turn, command tower, debug	
Skills	Computational thinking, measuring angles, teamwork, problem	

	solving
National Identity Integration	Link angles to the geometry in Emirati architecture: the patterns on the Al Fahidi wind towers and mosque windows are built from precise angles.
AI/Digital Learning	Matatalab Pro Sets; hook video; live quiz on the TCD website. Presentation: Grade4_Week10_Presentation.pptx. Worksheet: Grade4_Week10_Worksheet.

Lesson Structure	
Starter (Hook) Engage and Ask (10 minutes)	Warm-up: Human Robot (slide 4). Show the hook video and ask the hook question (slide 2). Share the learning goals (slide 3) and key words (slide 5).
I Do (20 minutes)	Refresh the coding blocks and the Command Tower (slide 6). Teach degrees with a protractor and body turns (slide 7). Model using an angle block to draw a square on A3 paper (slides 8 and 9).
We Do (15 minutes)	Pairs run a short refresh route on the map mat and fix one bug together. Class plans the triangle code together on the board: how many turns, and how big?
You Do (35 minutes)	Pairs complete the shape challenge and record their code on the worksheet (slide 11). Challenge levels on the board (slide 12). Live quiz on tablets (slide 13).
Differentiated Instruction: **Beginning / On / Above**	**Beginning: Draw the square only, using a ready-made code card to copy from.** **On: Draw the square and the triangle, planning the code yourself.** **Above: Draw a hexagon (60 degree turns) or a star and explain the turn you used.**
Closure/ Reflection (10 minutes)	Two pairs show their drawings and explain their turn angles. Exit ticket: one thing an angle block does (slide 14). Pack away sets and check pens are capped.

Inclusive Planning
Students receiving additional inclusive support (IEP or ILPs) or Gifted and Talented: EAL: block picture cards and the word wall. SEN: ready-made code cards to copy; mixed-ability pairs. G&T: hexagon or star challenge with a written explanation of the turn angle.
Inclusion / EAL Support

☑ Visual Supports
☐ Chunked Instructions
☐ Reduced Language Load
☑ Model / Worked Example
☐ Pre-teach Vocabulary
☐ TA/LSA Support
☐ Extra Processing Time
☐ Text Read by TA/LSA
☑ Flexible Grouping
☐ Assistive Technology
☐ Movement Break
☑ Scaffold / Sentence Starters
☐ Other:

G&T / Extension

☑ Extension / Challenge Task
☑ Higher-order Questioning
☐ Depth & Complexity Task
☐ Independent Inquiry
☐ Leadership / Peer Coaching
☐ Real-World Application
☐ Curriculum Compacting
☐ Above-grade-level Texts
☐ Student-led / Passion Project
☑ Open-Ended Task
☐ Other:

Assessment Adaptations

☑ Oral Response
☐ Reduced Questions
☑ Scaffolded Success Criteria
☐ Extra Time
☐ Alternative Outcome Format
☑ Teacher Conferencing/Live Feedback
☐ Quiet / Separate Space
☐ Other: